

Attention

Attention Mechanism in Deep Learning

Overview:

The attention mechanism allows models to focus on the most relevant parts of input when producing each output. It computes a set of weights that represent the importance of input elements relative to a given query.

Key Concepts:

- Query, Key, Value: Inputs are projected into queries (Q), keys (K), and values (V). Attention scores are computed from Q and K, then applied to V.
- Scaled Dot-Product Attention: $\text{scores} = \text{softmax}((Q K^T) / \sqrt{d_k}) V$. Scaling by $\sqrt{d_k}$ stabilizes gradients.
- Multi-Head Attention: multiple attention heads run in parallel with different learned projections, then concatenated. This allows the model to capture different types of relationships.
- Self-Attention: Q, K, V all come from the same source (e.g., tokens in a sentence). Enables contextualized representations.

Benefits:

- Captures long-range dependencies more effectively than RNNs.
- Parallelizable computations (especially in Transformers).
- Improves interpretability via attention weights (though weights are not perfect explanations).

Applications:

- Machine translation, summarization, question answering, language modeling (Transformers like BERT, GPT).
- Vision: visual attention and Vision Transformers (ViT).

Variants and Extensions:

- Additive (Bahdanau) attention: uses a small feed-forward network to compute scores.
- Relative positional encodings, sparse attention, cross-attention between modalities.

Practical Tips:

- Use layer normalization and residual connections with attention layers.
- Tune head count and model dimension; more heads can help but increase compute.
- Combine attention with positional encodings to capture order information.

Further Reading:

- "Attention Is All You Need" (Vaswani et al., 2017) — foundational Transformer paper.
- Tutorials on scaled dot-product and multi-head attention.

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